INCH-POUND

MIL-B-18/10E
w/AMENDMENT 1
5 February 2021
SUPERSEDING
MIL-B-18/10E
1 March 2011

MILITARY SPECIFICATION SHEET

BATTERY, DRY, BA-31

INACTIVE FOR NEW DESIGN AFTER 22 JULY 1998

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-B-18.

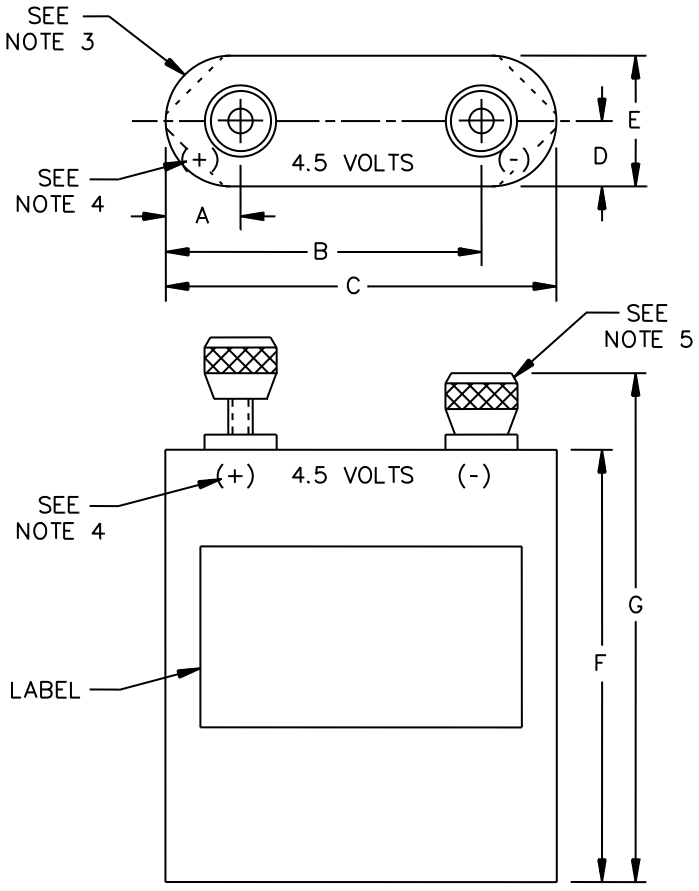


FIGURE 1. Interface and physical dimensions.

AMSC N/A

FSC 6135

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:40Z
Check the source to verify that this is the current version before use.

Ltr	Inches		mm	
	Min	Max	Min	Max
A	.406	.532	10.32	13.51
B	1.906	2.032	48.41	51.61
C	2.344	2.532	59.54	64.31
D	.343	.469	8.71	11.91
E	.750	.874	19.05	22.20
F	2.625	2.751	66.68	69.88
G	---	3.156	---	80.16

NOTES:
1. All dimensions shown on figure are in inches.
2. Metric equivalents specified in table are for general information only.
3. Diagonal or round corners are optional.
4. Voltage and polarity markings may be placed either on the top or front face of the battery.
5. Unless otherwise specified, tolerance is ±.062 inch.
6. Provide insulated knurled nuts for brass inserts.

FIGURE 1. Interface and physical dimensions - Continued.

REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Nominal voltage: 4.5 volts.

Usual number and type of cells: 3 "B" cells.

Usual cell connection: Series

Terminals: Stud and nut.

Weight (maximum): 6 ounces.

Capacity tests: When the battery is tested in accordance with the methods of examination and test of this specification, the minimum capacity-test requirements shall be not less than the minimum time specified for SLD or SLT.

First article inspection:

Visual and mechanical (external).

Battery voltage.

Vibration test.

Mechanical-shock test in accordance with MIL-STD-202-213, test condition I.

Insulation resistance test in accordance with MIL-STD-202-302, test condition B, tolerance ± 20 volts.

Capacity, D (without storage).

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:40Z
Check the source to verify that this is the current version before use.

Jacket integrity test.

Conformance inspection.

Visual and mechanical (external)

Battery voltage.

Insulation-resistance test in accordance with MIL-STD-202-302, test condition B, tolerance ± 20 volts.

Vibration test.

Mechanical-shock test in accordance with MIL-STD-202-213, test condition I.

Jacket integrity test.

Electrode leakage.

Capacity, D and T.

Methods of examination and test:

Capacity tests: See requirements for capacity specified herein.

Storage:	Test	Period
	D	12 months
	T	90 days

Open-circuit voltage: After completion of storage for D test specified, the open circuit voltage shall be not less than 4.35 volts when measured at ambient storage conditions.

Discharge: The battery shall be discharged through 20 ohms for 4 minutes per hour, 10 hours a day, 5 days per week. This cycle shall be repeated continuously to a test-end voltage of 2.8 volts.

Closed circuit voltage: Use TS-183( )/U test per special marking or, with minimum permissible voltage as specified for TS-183 ( )/U, use a load resistant value of 19.95 ohms.

Special marking on each unit package:

TO TEST THIS BATTERY WITH TS-183 ( )/U:

USE JACK NO	MINIMUM PERMISSIBLE VOLTAGE
7	3,84

Unless expressly authorized, this test information shall apply to Army and Air Force applications only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:40Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN): BA-31, see 1.2 of MIL-B-18.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-B-18, this document references the following:

MIL-STD-202-213 MIL-STD-202-302

Custodians:
Army - CR
Navy - SH
DLA - CC

Review activities:
Army - CR4, MI
Navy - MC

Preparing activity:
DLA - CC

(Project 6135-2020-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T16:40Z
Check the source to verify that this is the current version before use.